Genome-Wide Association Exercise Association Analysis Controlling for Population Substructure

1. Population Stratification and Association Testing

The dataset from part I of this exercise which you performed data quality control (QC) on was obtained from HapMap Phase III data. It contains CEU founders (Caucasians from Utah), MEX founders (Mexicans from Los Angeles) and TSI (Tuscans from Italy). The CEU pedigree identifiers begin with only numbers e.g., 1347, the MEX pedigree identifies all start with M e.g., M017 and the TSI pedigree identifiers all start with NA e.g., NA0217. Before we start testing for association, we want to know if there are outliers. Even after removing the outliers when association analysis is performed population substructure and admixture may need to be controlled. If not, we risk observing an association, which is due to a difference in genotype frequencies in cases and controls, because of population substructure/admixture and not because of linkage disequilibrium (LD) between tagSNP(s) and the functional variant(s). We are going to use multidimensional scaling (MDS) and principal components analysis (PCA) within the PLINK software to generate 10 components.

Disclaimer: You usually should not analyze data from European-Americans, Mexican- Americans and Italians together even if you control for population stratification. They can be analyzed separately, and the data combined using meta-analysis.

Note: For a GWAS study instead of this toy study, you will have a denser set of markers of which some will be in LD. You should first prune your SNPs to obtain a subset in linkage equilibrium/weak LD (R2<0.1) prior to performing MDS or PCA analysis on the data. Although for association analysis is performed on the entire data set will be analyzed only this a subset of SNPs which are not in LD will be used to construct PCA and MDS components. For more information on how to do this in PLINK see https://www.cog-genomics.org/plink/1.9/ld.

A note on running PLINK from this notebook

This notebook runs in a single R kernel from start to finish — no more switching between a Bash kernel and an R kernel. PLINK is a command-line program, not an R package, so to run it from R we use the built-in `system()` function. `system()` hands its argument (a string) to your computer's shell exactly as if you had typed it into a terminal yourself, waits for it to finish, and lets any files it writes (`.log`, `.mds`, `.eigenvec`, etc.) show up in your working directory as usual.

For example, the R code

```
system("plink --file GWAS_clean4 --genome --cluster --mds-plot 10")
```

does exactly the same thing as opening a terminal, `cd`-ing into the data directory, and typing:

```
plink --file GWAS_clean4 --genome --cluster --mds-plot 10
```

Every `system()` call below is preceded by a comment showing that plain command-line form, so you can always see exactly what is being run — you could copy that line straight into a terminal and get the same result.

Set Kernel to R

In [1]:

```r
# Move into the working directory containing the PLINK input files.
# Command-line equivalent:
#   cd ./data
setwd("./data/")
```

In [2]:

```r
# Command-line equivalent:
#   plink --file GWAS_clean4 --genome --cluster --mds-plot 10
plink_out <- system("plink --file GWAS_clean4 --genome --cluster --mds-plot
cat(plink_out, sep = "\n")
```

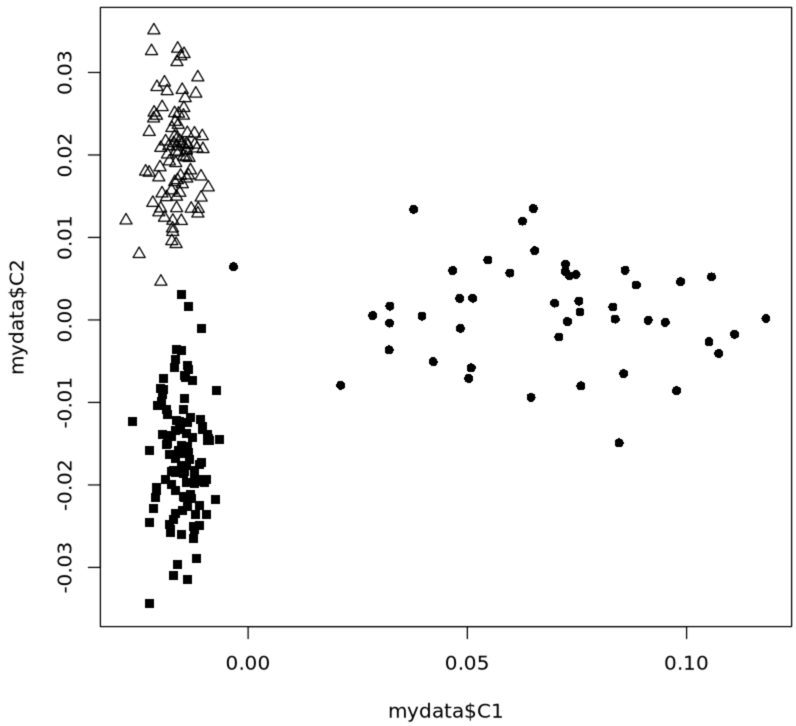
```
PLINK v1.9.0-b.8 64-bit (22 Oct 2024)            cog-genomics.org/plink/1.
9/
(C) 2005-2024 Shaun Purcell, Christopher Chang   GNU General Public License
v3
Logging to plink.log.
Options in effect:
  --cluster
  --file GWAS_clean4
  --genome
  --mds-plot 10

7836 MB RAM detected; reserving 3918 MB for main workspace.
.ped scan complete (for binary autoconversion).
Performing single-pass .bed write (6362 variants, 244 people).
--file: plink-temporary.bed + plink-temporary.bim + plink-temporary.fam
written.
6362 variants loaded from .bim file.
244 people (123 males, 121 females) loaded from .fam.
Using up to 9 threads (change this with --threads).
Before main variant filters, 244 founders and 0 nonfounders present.
Calculating allele frequencies... done.
Warning: 6 het. haploid genotypes present (see plink.hh ); many commands tre
at
these as missing.
Total genotyping rate is 0.997229.
6362 variants and 244 people pass filters and QC.
Note: No phenotypes present.
Excluding 186 variants on non-autosomes from IBD calculation.
IBD calculations complete.
Finished writing plink.genome .
Clustering... done.
Cluster solution written to plink.cluster1 , plink.cluster2 , and
plink.cluster3 .
Performing multidimensional scaling analysis (SVD algorithm, 10
dimensions)... done.
MDS solution written to plink.mds .
```

This command outputs the file `plink.mds` that contains the subject IDs and values for the 10 components we just generated. There is another file in your folder called `mds_components.txt`. This file is identical to your `plink.mds` file with the exception that a group column which codes CEU individuals as 1, MEX individuals as 2 and TSI individuals as 3. This is done so when we plot the MDS components in R you can see which group the points belong to and judge how well does the data cluster, e.g., are there outliers. The following commands will generate a jpeg image file containing the mds `plot (filename=mds.jpeg)` in your current working directory. Open R and use the following command:

In [3]:
```
mydata = read.table("mds_components.txt", header=T)
mydata$pch[mydata$Group==1 ] <-15
mydata$pch[mydata$Group==2 ] <-16
mydata$pch[mydata$Group==3 ] <-2
#jpeg("mds.jpeg", height=500, width=500)
```

```
plot(mydata$C1, mydata$C2 ,pch=mydata$pch)
dev.off()
```

null device: 1



Visualizing population structure using MDS is useful for identifying subpopulations, population stratification and systematic genotyping or sequencing errors, and can also be used to detect individual outliers that may need to be removed, e.g. European-Americans included in a study of African-Americans. MDS coordinates help with visualizing genetic distances and population substructure. PLINK also offers another dimension reduction, --pca, for PCA, the PC components which can also be used for visualizing data to detect outliers in the same manner which was performed using MDS. Additionally, covariates either from either MDS or PCA can be used in a regression model to aid in correcting for population substructure and admixture.

We will now continue performing the analysis using PLINK but will use PCA instead of MDS. We will generate PCs and determine how many PC covariates should be included in the regression model. When SNPs are tested for an association with a trait analysis can be performed, first by including no PC components, then one PC component and then two PC components and so on. Please note that as each PC component is added all the SNPs are analyzed, e.g. a complete GWAS is performed. Examining λ can aid in determining how many PC components should be included in the analysis. If there is no population stratification or other biases, then λ should equal 1 or ~1. We will use λ to determine how many PC components from our analysis will be added to the logistic regression model. First, estimate λ without adjusting for any PC components:

In [4]:
```r
# Command-line equivalent:
#   plink --file GWAS_clean4 --pheno pheno.txt --pheno-name Aff --logistic -
plink_out <- system("plink --file GWAS_clean4 --pheno pheno.txt --pheno-name
cat(plink_out, sep = "\n")
```

```
PLINK v1.9.0-b.8 64-bit (22 Oct 2024)              cog-genomics.org/plink/1.
9/
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v3
Logging to unadj.log.
Options in effect:
  --adjust
  --file GWAS_clean4
  --logistic
  --out unadj
  --pheno pheno.txt
  --pheno-name Aff

7836 MB RAM detected; reserving 3918 MB for main workspace.
.ped scan complete (for binary autoconversion).
Performing single-pass .bed write (6362 variants, 244 people).
--file: unadj-temporary.bed + unadj-temporary.bim + unadj-temporary.fam
written.
6362 variants loaded from .bim file.
244 people (123 males, 121 females) loaded from .fam.
244 phenotype values present after --pheno.
Using 1 thread (no multithreaded calculations invoked).
Before main variant filters, 244 founders and 0 nonfounders present.
Calculating allele frequencies... done.
Warning: 6 het. haploid genotypes present (see unadj.hh ); many commands tre
at
these as missing.
Total genotyping rate is 0.997229.
6362 variants and 244 people pass filters and QC.
Among remaining phenotypes, 87 are cases and 157 are controls.
Writing logistic model association results to unadj.assoc.logistic ... done.
--adjust: Genomic inflation est. lambda (based on median chisq) = 1.12068.
--adjust values (6346 variants) written to unadj.assoc.logistic.adjusted .
```

Generated the first 10 PCA values:

In [5]:
```r
# Command-line equivalent:
#   plink --file GWAS_clean4 --genome --cluster --pca 10 header
plink_out <- system("plink --file GWAS_clean4 --genome --cluster --pca 10 he
cat(plink_out, sep = "\n")
```

```
PLINK v1.9.0-b.8 64-bit (22 Oct 2024)            cog-genomics.org/plink/1.
9/
(C) 2005-2024 Shaun Purcell, Christopher Chang   GNU General Public License
v3
Logging to plink.log.
Options in effect:
  --cluster
  --file GWAS_clean4
  --genome
  --pca 10 header

7836 MB RAM detected; reserving 3918 MB for main workspace.
.ped scan complete (for binary autoconversion).
Performing single-pass .bed write (6362 variants, 244 people).
--file: plink-temporary.bed + plink-temporary.bim + plink-temporary.fam
written.
6362 variants loaded from .bim file.
244 people (123 males, 121 females) loaded from .fam.
Using up to 9 threads (change this with --threads).
Before main variant filters, 244 founders and 0 nonfounders present.
Calculating allele frequencies... done.
Warning: 6 het. haploid genotypes present (see plink.hh ); many commands tre
at
these as missing.
Total genotyping rate is 0.997229.
6362 variants and 244 people pass filters and QC.
Note: No phenotypes present.
Excluding 186 variants on non-autosomes from relationship matrix calc.
Relationship matrix calculation complete.
--pca: Results saved to plink.eigenval and plink.eigenvec .
Excluding 186 variants on non-autosomes from IBD calculation.
IBD calculations complete.
Finished writing plink.genome .
Clustering... done.
Cluster solution written to plink.cluster1 , plink.cluster2 , and
plink.cluster3 .
```

Eigenvectors are written to plink.eigenvec, and top eigenvalues are written to plink.eigenval. The 'header' modifier adds a header line to the .eigenvec file(s). And then find out what λ is when we adjust for the first component:

In [6]:

```
# Command-line equivalent:
#   plink --file GWAS_clean4 --pheno pheno.txt --pheno-name Aff --covar plir
plink_out <- system("plink --file GWAS_clean4 --pheno pheno.txt --pheno-name
cat(plink_out, sep = "\n")
```

```
PLINK v1.9.0-b.8 64-bit (22 Oct 2024)            cog-genomics.org/plink/1.
9/
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v3
Logging to PC1.log.
Options in effect:
  --adjust
  --covar plink.eigenvec
  --covar-name PC1
  --file GWAS_clean4
  --logistic
  --out PC1
  --pheno pheno.txt
  --pheno-name Aff

7836 MB RAM detected; reserving 3918 MB for main workspace.
.ped scan complete (for binary autoconversion).
Performing single-pass .bed write (6362 variants, 244 people).
--file: PC1-temporary.bed + PC1-temporary.bim + PC1-temporary.fam written.
6362 variants loaded from .bim file.
244 people (123 males, 121 females) loaded from .fam.
244 phenotype values present after --pheno.
Using 1 thread (no multithreaded calculations invoked).
--covar: 1 out of 10 covariates loaded.
Before main variant filters, 244 founders and 0 nonfounders present.
Calculating allele frequencies... done.
Warning: 6 het. haploid genotypes present (see PC1.hh ); many commands treat
these as missing.
Total genotyping rate is 0.997229.
6362 variants and 244 people pass filters and QC.
Among remaining phenotypes, 87 are cases and 157 are controls.
Writing logistic model association results to PC1.assoc.logistic ... done.
--adjust: Genomic inflation est. lambda (based on median chisq) = 1.08462.
--adjust values (6346 variants) written to PC1.assoc.logistic.adjusted .
```

And the first and second components:

In [7]:

```
# Command-line equivalent:
#   plink --file GWAS_clean4 --pheno pheno.txt --pheno-name Aff --covar plir
plink_out <- system("plink --file GWAS_clean4 --pheno pheno.txt --pheno-name
cat(plink_out, sep = "\n")
```

```
PLINK v1.9.0-b.8 64-bit (22 Oct 2024)            cog-genomics.org/plink/1.
9/
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v3
Logging to PC1-PC2.log.
Options in effect:
  --adjust
  --covar plink.eigenvec
  --covar-name PC1-PC2
  --file GWAS_clean4
  --logistic
  --out PC1-PC2
  --pheno pheno.txt
  --pheno-name Aff

7836 MB RAM detected; reserving 3918 MB for main workspace.
.ped scan complete (for binary autoconversion).
Performing single-pass .bed write (6362 variants, 244 people).
--file: PC1-PC2-temporary.bed + PC1-PC2-temporary.bim + PC1-PC2-temporary.fa
m
written.
6362 variants loaded from .bim file.
244 people (123 males, 121 females) loaded from .fam.
244 phenotype values present after --pheno.
Using 1 thread (no multithreaded calculations invoked).
--covar: 2 out of 10 covariates loaded.
Before main variant filters, 244 founders and 0 nonfounders present.
Calculating allele frequencies... done.
Warning: 6 het. haploid genotypes present (see PC1-PC2.hh ); many commands
treat these as missing.
Total genotyping rate is 0.997229.
6362 variants and 244 people pass filters and QC.
Among remaining phenotypes, 87 are cases and 157 are controls.
Writing logistic model association results to PC1-PC2.assoc.logistic ... don
e.
--adjust: Genomic inflation est. lambda (based on median chisq) = 1.02628.
--adjust values (6346 variants) written to PC1-PC2.assoc.logistic.adjusted .
```

and so forth for all 10 components in the .log file completing the table:

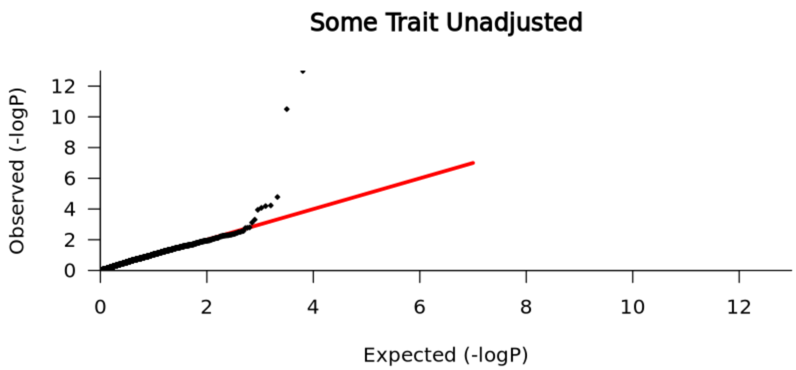
The number closest to 1.0, with the least number of PC components, would be the best for adjusting without overfitting and introducing unnecessary noise. You can check your table against the one provided in the answers section.

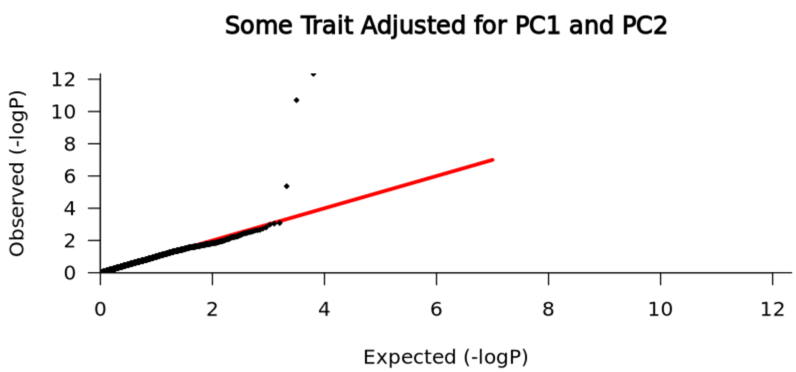
Go to the `assoc.logistic` **file that corresponds to that number of components** and make a note of how you named the `.assoc.logistic` file for it and when you did not adjust for any components. Then go back to the R program to load the results and create a jpeg image file containing QQ plots for the adjusted and unadjusted results (using a modified script from http://www.broad.mit.edu/node/555) as follows:

In [8]:
```
broadqq <-function(pvals, title) {
    observed <- sort(pvals)
    lobs <- -(log10(observed))
```

```
    expected <- c(1:length(observed))
    lexp <- -(log10(expected / (length(expected)+1)))
    plot(c(0,7), c(0,7), col="red", lwd=3, type="l", xlab="Expected (-logP)"
         xlim=c(0,max(lobs)), ylim=c(0,max(lobs)), las=1, xaxs="i", yaxs="i"
    points(lexp, lobs, pch=23, cex=.4, bg="black")
}
#jpeg("qqplot_compare.jpeg", height=1000, width=500)
par(mfrow=c(2,1))
aff_unadj<-read.table("unadj.assoc.logistic", header=TRUE)
aff_unadj.add.p<-aff_unadj[aff_unadj$TEST==c("ADD"),]$P
broadqq(aff_unadj.add.p,"Some Trait Unadjusted")
aff_C1C2<-read.table("PC1-PC2.assoc.logistic", header=TRUE)
aff_C1C2.add.p<-aff_C1C2[aff_C1C2$TEST==c("ADD"),]$P
broadqq(aff_C1C2.add.p, "Some Trait Adjusted for PC1 and PC2")
dev.off()
```

null device: 1





Now look for SNPs with genome-wide significance using the following R connamds:

In [9]:
```
gws_unadj = aff_unadj[which(aff_unadj$P < 0.0000001),]
gws_unadj
gws_adjusted = aff_C1C2[which(aff_C1C2$P < 0.0000001),]
gws_adjusted
```

A data.frame: 2 × 9

	CHR	SNP	BP	A1	TEST	NMISS	OR	STAT	P
	<int>	<chr>	<int>	<chr>	<chr>	<int>	<dbl>	<dbl>	<dbl>
1800	4	rs10008252	179853616	G	ADD	244	0.16650	-6.639	3.155e-11
3131	8	rs4571722	60326734	T	ADD	242	0.04126	-7.436	1.037e-13

A data.frame: 2 × 9

	CHR	SNP	BP	A1	TEST	NMISS	OR	STAT	P
	<int>	<chr>	<int>	<chr>	<chr>	<int>	<dbl>	<dbl>	<dbl>
5398	4	rs10008252	179853616	G	ADD	244	0.13070	-6.707	1.985e-11
9391	8	rs4571722	60326734	T	ADD	242	0.04382	-7.237	4.585e-13

Note: These are the uncorrected p-values for multiple testing. The p-values which have been corrected using various multiple testing methods can be found in the .adjusted file.

A common question when you have a finding with genome-wide significance in a GWAS is “Is the SNP in a known gene?” One way to look this information up is annotate variants in batch (please look at the annotating exercise for more information). You can do this using the Ensembl Variant Predictor. Go to the website:

http://grch37.ensembl.org/Homo_sapiens/Tools/VEP (GRCh37 version)

Type the rs number(s) of the SNP(s) with genome-wide significance in “Either paste data”, leave all options default and press run. In a few minutes you can view the results of your query.